

Manufacturing & Supplier Quality

Medical Early Warning & Patient Monitoring Platform
Portfolio Reference Document | Prepared by Christopher Anthony Agustin | Senior Project Manager

1. Purpose

This reference describes the manufacturing-management approach used to move product development toward pilot production and production readiness. It focuses on supplier coordination, quality responsibilities, manufacturing controls, traceability, process changes, inspection, and readiness.

2. Manufacturing Strategy

Stage	Management Focus	Expected Output
Supplier Selection & Qualification	Evaluate capability, quality expectations, responsibilities, and readiness	Qualified manufacturing partner
Product Transfer	Transfer drawings, specifications, component information, and work instructions	Production-ready documentation set
Pilot Build	Coordinate initial build, inspection, testing, and issue resolution	Pilot build evidence and action list
Mass Production Readiness	Confirm process controls, inspection, packaging, traceability, and capacity	Production readiness decision
Ongoing Monitoring	Review quality performance, changes, nonconformities, and corrective actions	Continuous improvement and supplier oversight

3. Supplier Quality Management

The supplied reference agreement shows a formal supplier-quality model: quality responsibilities are defined between the parties; materials, components, and products are expected to meet agreed specifications and applicable requirements; supplier quality systems, inspection, traceability, audits, process changes, and nonconforming products are addressed.

Control	Manufacturing Application
Supplier Quality System	Supplier maintains an effective quality system and defined inspection activities.
Incoming / In-Process / Final Inspection	Materials and products are checked against agreed specifications before progressing.
Traceability	Supplied products and related records remain identifiable through the manufacturing process.
Change Control	Changes to process, materials, equipment, or manufacturing location are reviewed before implementation.
Audit Rights	Manufacturing facilities, records, and relevant processes may be subject to supplier audits / inspections.
Nonconformance	Nonconforming products are reported and corrective actions are initiated.

4. Manufacturing Process Controls

Process Area	Key Controls
Material Receipt	Verify supplier deliveries, identity, condition, and specification compliance.
Assembly	Use defined work instructions, tooling, materials, and trained personnel.
In-Process Inspection	Identify defects early and prevent nonconforming work from moving downstream.
Verification & Testing	Confirm functional and product requirements before final acceptance.
Final Inspection	Review finished product against defined acceptance criteria.
Packaging & Shipping	Maintain labeling, serialization, traceability, and shipment records.

5. PM Responsibilities

- Coordinate engineering, quality, supplier, manufacturing, and project priorities.

- Maintain visibility of production dependencies, risks, issues, and readiness criteria.
- Track technical and process changes that may affect product quality or downstream activities.
- Facilitate communication between internal stakeholders and manufacturing partners.
- Ensure unresolved quality issues and corrective actions remain visible through closure.

6. Manufacturing Readiness Checklist

Readiness Area	Checkpoint
Supplier	Qualified supplier and quality responsibilities agreed
Documentation	Current drawings, specifications, work instructions, and process information available
Materials	Required components and materials available and traceable
Process	Assembly and inspection activities defined and understood
Testing	Required verification / inspection activities identified
Quality	Nonconformance and corrective-action process established
Change Control	Manufacturing changes have defined review and approval path
Traceability	Product / component records can be traced through the process
Packaging	Labeling, serialization, packaging, and shipment controls defined
Production	Pilot results reviewed and production readiness assessed

7. Risk & Issue Management

Risk / Issue	Management Response
Supplier quality issue	Contain affected material / product, assess impact, assign corrective action, and monitor closure.
Process change	Assess impact on quality and downstream validation before implementation.
Component availability	Monitor supply dependencies and coordinate alternatives through controlled change processes.
Production defect	Record the nonconformance, determine disposition, identify cause, and track corrective action.
Readiness gap	Identify owner, due date, dependency, and decision required before production progression.

8. PM Outcome

A structured manufacturing approach connects supplier management, production controls, quality responsibilities, traceability, change management, and readiness decisions so that product development can transition into repeatable production.

Portfolio Reference Document - based on project documentation and supplier-quality evidence. It intentionally omits confidential company and supplier identifiers.